

Statistical Modeling of Vehicle Fuel Efficiency: Regression Analysis of the Auto MPG Dataset

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Abstract

This report investigates the determinants of automobile fuel economy (MPG) using the Auto dataset. We utilize Simple Linear Regression (SLR) and Multiple Linear Regression (MLR) to identify key predictors, evaluate interaction effects, and explore non-linear transformations to optimize model performance.

1 Theoretical Foundations

1.1 Advertising Channel Significance (Conceptual)

In a multiple regression context, the p -value for a specific coefficient tests the null hypothesis (H_0) that the predictor has no effect on the response, provided all other predictors are held constant.

- **TV & Radio:** Both show $p < 0.0001$, suggesting significant independent contributions to sales.
- **Newspaper:** With $p = 0.8599$, it fails to show a significant marginal effect when TV and Radio budgets are already accounted for, suggesting its impact might be redundant or captured by other variables.

1.2 The Hat Matrix and Leverage

For a regression model without an intercept, the predicted values $\hat{y}_i$ are derived from $x_i\hat{\beta}$. The derivation follows:

$$\hat{\beta} = \frac{\sum_{i'=1}^n x_{i'} y_{i'}}{\sum_{i'=1}^n x_{i'}^2}$$

Substituting $\hat{\beta}$ into the prediction equation:

$$\hat{y}_i = x_i \left(\frac{\sum_{i'=1}^n x_{i'} y_{i'}}{\sum_{j=1}^n x_j^2} \right) = \sum_{i'=1}^n \left(\frac{x_i x_{i'}}{\sum_{j=1}^n x_j^2} \right) y_{i'}$$

This demonstrates that the weight of the i' -th observation on the i -th prediction is given by $a_{ii'} = \frac{x_i x_{i'}}{\sum_{j=1}^n x_j^2}$.

2 Univariate Analysis: The Impact of Horsepower

We initially modeled fuel efficiency as a linear function of engine power: $\text{mpg} = \beta_0 + \beta_1(\text{horsepower}) + \epsilon$.

2.1 Empirical Results

The OLS estimation yielded a baseline MPG of 39.94 with a significant decay rate of -0.1578 per unit of horsepower.

Metric	Value	Stat. Significance	Interpretation
Coefficient ($\hat{\beta}_1$)	-0.1578	$p \approx 0$	Strong Inverse Relation
R^2	0.6059	—	60.6% Variance Explained
Prediction (HP=98)	24.47	95% CI: [23.97, 24.96]	Reliable Estimate

Table 1: Summary of Simple Linear Regression findings.

2.2 Diagnostics

As shown in Figure 1, the *Residuals vs Fitted* plot reveals a distinct parabolic trend. This suggests that the relationship between horsepower and fuel economy is inherently non-linear, warranting a more complex model or transformations.

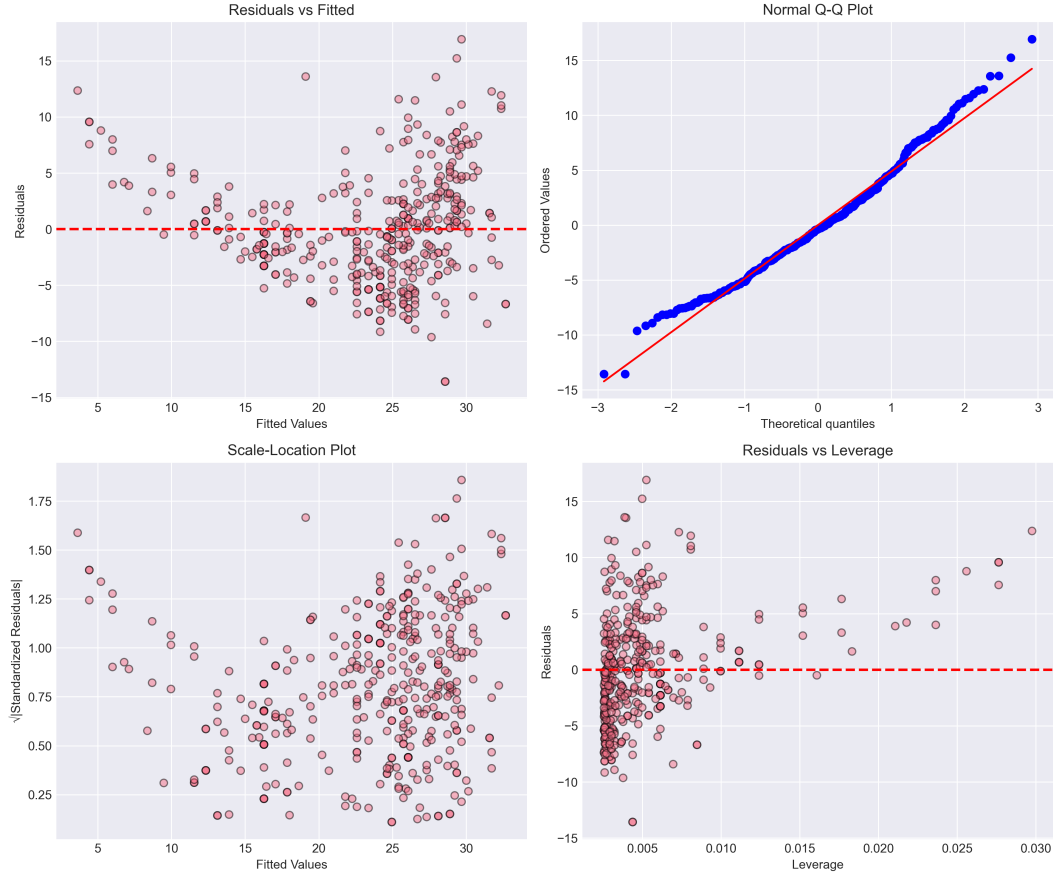


Figure 1: SLR Diagnostics: Note the non-linearity in residuals.

3 Multivariate Modeling and Feature Engineering

3.1 Variable Selection and Multicollinearity

The scatterplot matrix and heatmap (Fig 2) confirm that predictors like *displacement*, *weight*, and *horsepower* are highly collinear ($r > 0.8$). This is further reflected in the high condition number (8.59×10^4), suggesting that individual coefficient estimates may be unstable despite a high overall R^2 of 0.8215.

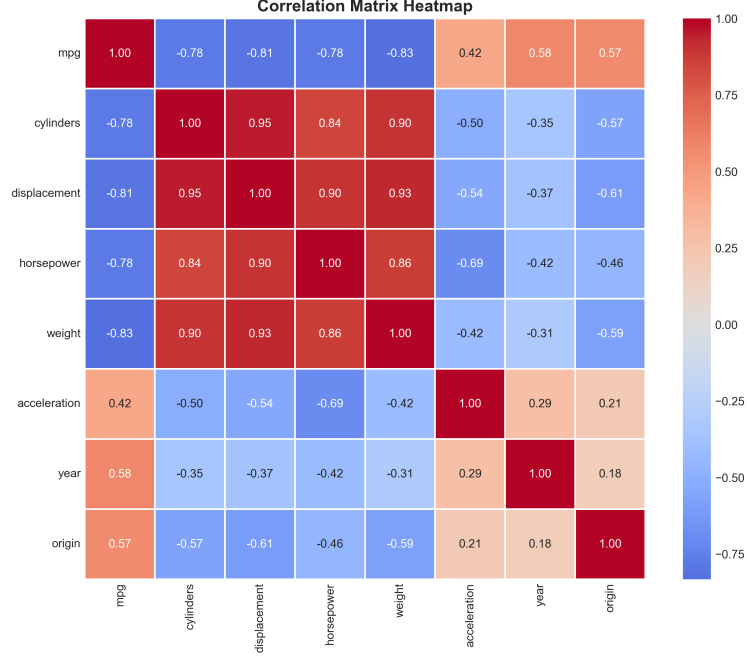


Figure 2: Correlation Analysis: Darker regions indicate strong collinearity.

3.2 Refined Model Insights

The MLR analysis identifies **weight**, **year**, **origin**, and **displacement** as the primary drivers of MPG. Interestingly, once weight and displacement are included, *cylinders* and *acceleration* lose their statistical significance ($p > 0.05$).

3.3 Interactions and Non-Linearity

To address model limitations, we tested interaction terms and logarithmic scales:

1. **Synergy:** The interaction between *weight* and *horsepower* is highly significant ($p < 0.001$), implying that the effect of weight on fuel economy changes as horsepower increases.
2. **Log-Transformation:** Applying a log scale to *weight* improved the model fit from $R^2 = 0.8215$ to 0.8459 . This logarithmic model better captures the diminishing returns of weight reduction on fuel efficiency.

4 Conclusion

While simple linear models provide a useful baseline, they fail to capture the curvilinear nature of automotive data. The transition from SLR to MLR, complemented by log-transformations and interaction terms, significantly enhances the predictive power. The final analysis confirms that vehicle weight remains the most critical factor, though its influence is modulated by engine technology (displacement) and manufacturing era (year).